

Course Code	EC24P
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Roll Number	23104B0003
Class	T.E – EXTC. SEM-V
Division	B
Date of Performance	25-7-25
Date of Submission	1-8-25

EXPERIMENT NO.2

Estimated time to complete this experiment: 02 hours.

Objective: To observe the effect of varying Aspect ratio of MOSFET on transfer and output characteristics of 180nm N channel MOSFET using CADENCE VIRTUOSO

CO to be achieved: CO1.

Expected Outcome of Experiment:

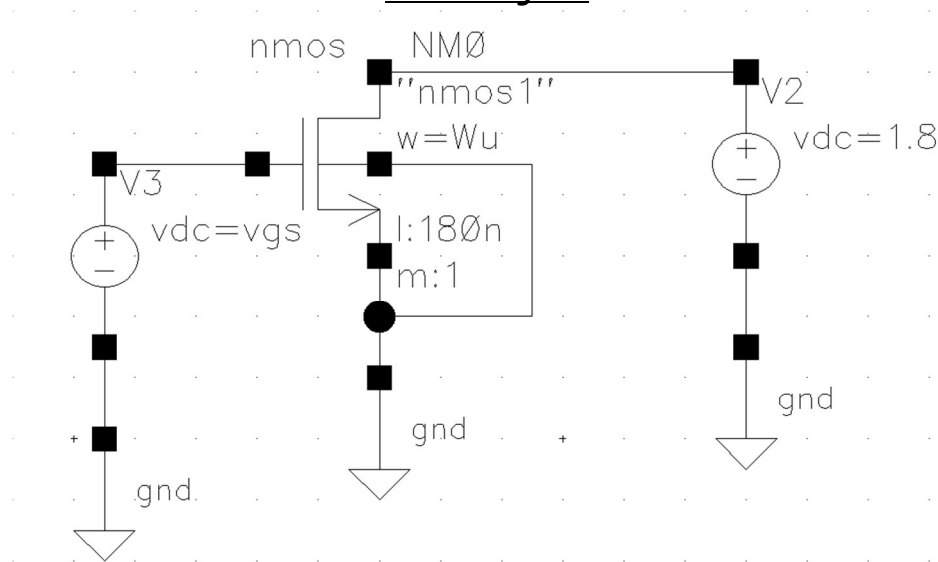
- Students will learn cadence virtuoso.
- To plot output and transfer characteristics of MOSFET for different values of Aspect ratio.
- To observe effect of varying aspect ratio on small signal parameters of MOSFET.

Pre Lab/ Prior Concept: Knowledge of working of MOSFET.

Apparatus/TOOL:

- PC
- CADENCE VIRTUOSO.

Circuit Diagram:



5) Procedure.

- Open the cadence virtuoso tool in Red-Hat Linux. The commands are
 - csh
 - source /home/install/cshrc
 - virtuoso.
- Create the Library and cell view
- Open log window using terminal.
- Click on the new option, now create library with gpdk180 from technology library.
- Again click the on new option, and set the cell view.
- Cadence Virtuoso dashboard will be opened.
- For characterization of 180nm NMOS Transistor
- Select NMOS transistor from add instance and select its channel length as 180nm (Nano meter) and channel width as Wu(micro-meter).
- Construct the circuit shown in circuit diagram in Cadence.
- Take VDD=1.8V and VGS=1V
- Then plot the Transfer and Output Characteristics, for W=1μM to 3 μM.
- From the transfer characteristics calculate gm of given NMOS Transistor for each value of width W.

$$g_m = \frac{ID_2 - ID_1}{V_{GS2} - V_{GS1}} \quad \text{or}$$

- From the output characteristics calculate ro for the given transistor for each value of width W.

$$r_o = \frac{V_{DS2} - V_{DS1}}{I_{D2} - I_{D1}} \Omega$$

- Now take $V_{DD}=1.8V$ $V_{GS}=1V$, channel width $W=1\mu m$ and Channel length= L nm(Nano-meter).
- Then plot the transfer and output characteristics for $L=180$ Nano-meter to 540 Nano-meter.
- From the transfer characteristics calculate g_m of given NMOS Transistor for each value of channel length L .

$$g_m = \frac{I_{D2} - I_{D1}}{V_{GS2} - V_{GS1}} \text{ } \cup$$

- From the output characteristics calculate r_o for the given transistor for each value of Channel length L .

$$r_o = \frac{V_{DS2} - V_{DS1}}{I_{D2} - I_{D1}} \Omega$$

down the value of drain current in saturation region and using that calculate the value of process Trans conductance $\mu_n C_{OX}$, using the drain current equation.

$$I_D = \frac{\mu_n C_{OX}}{2} * \left(\frac{W}{L}\right) * (V_{GS} - V_{TN})^2$$

- Using the value of r_o Calculate the value of λ (Channel length modulation coefficient) for each value of channel width W and channel length L .

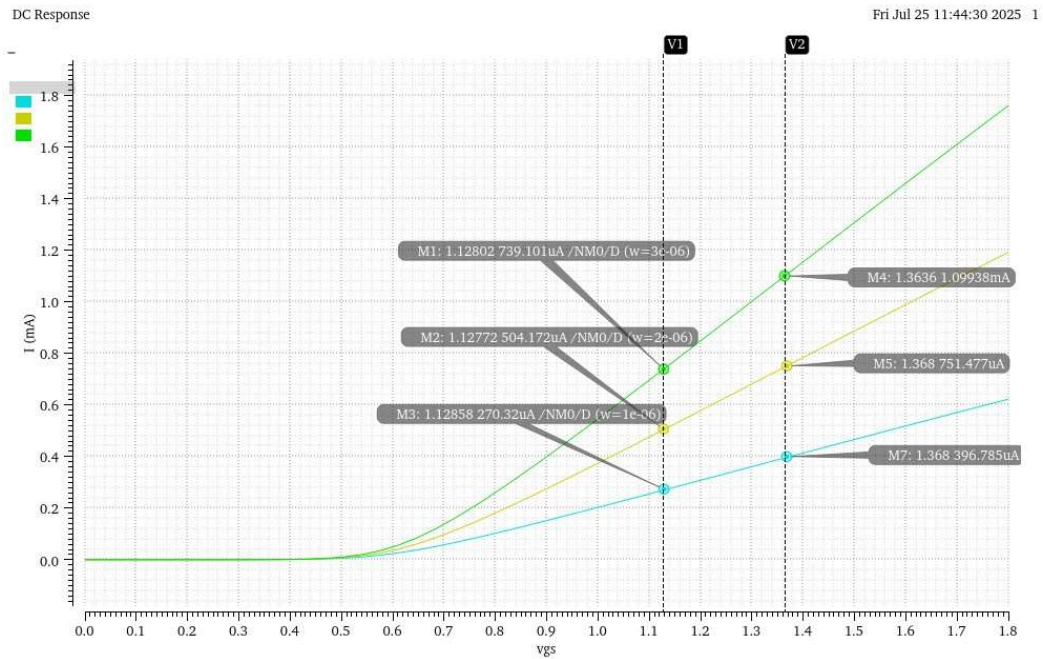
$$r_o = \frac{1}{\lambda * I_D} \Omega$$

- Finally calculate the Intrinsic gain of MOSFET= $g_m * R_o$
For each value of channel width W and channel length L .

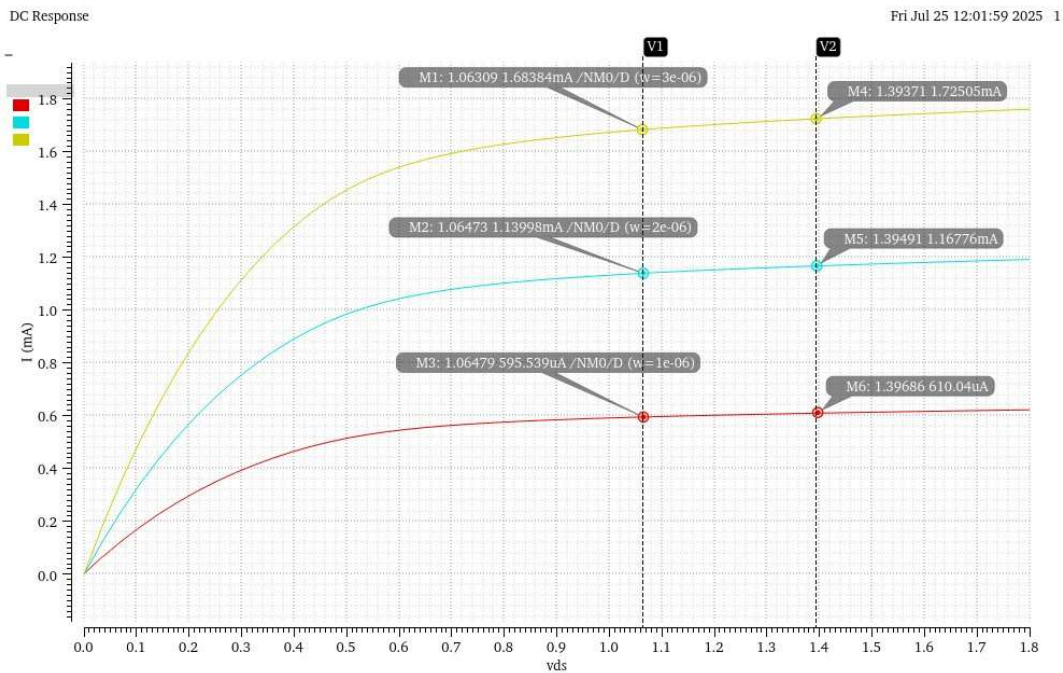
Results:

1.Varying Width

Transfer characteristics:

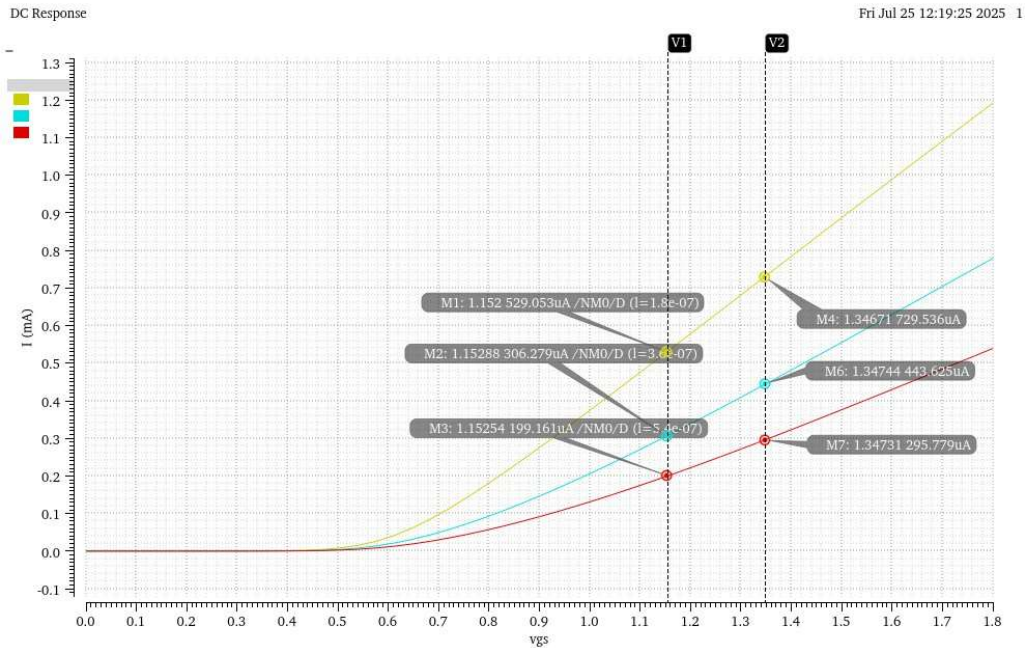


Output characteristics:

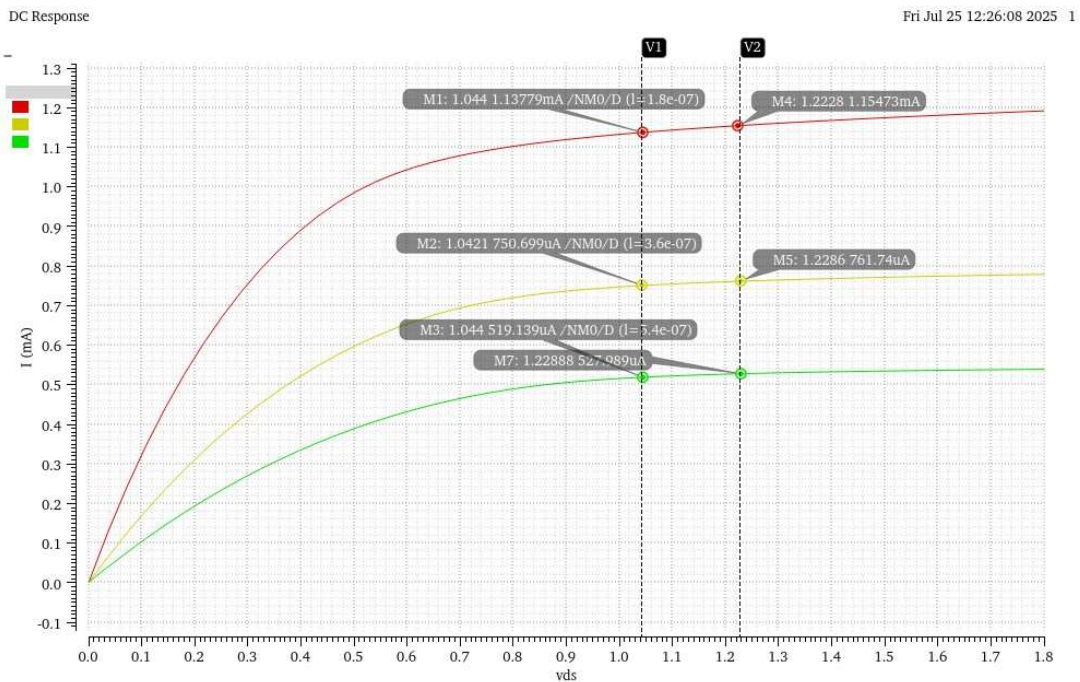


2.Varying Length

Transfer characteristics:



Output Characteristics:



Observation Table

Channel Width (W)	$I_{D_{SAT}}$	gm	r_o	λ	Intrinsic Gain
1 μ M	0.0005955	0.000527	22750	0.0074	11.98
2 μ M	0.00113399	0.001031	11877	0.0074	12.25
3 μ M	0.0016838	0.001496	8009	0.0074	11.98
Channel Length (L)	$I_{D_{SAT}}$	gm	r_o	λ	Intrinsic Gain
180nM	0.00113779	0.001028	10576	0.0835	10.86
360nM	0.000750699	0.000704	16329	0.0813	11.49
540nM	0.000519139	0.000497	20769	0.922	10.32

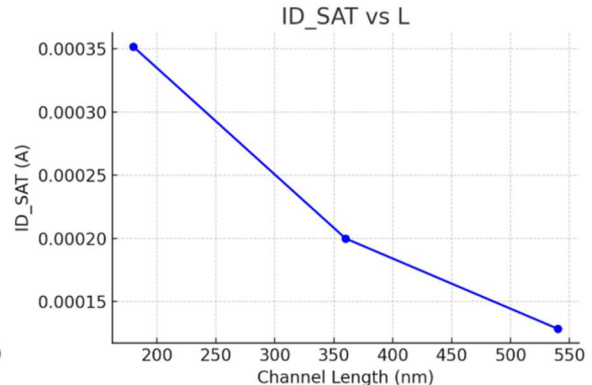
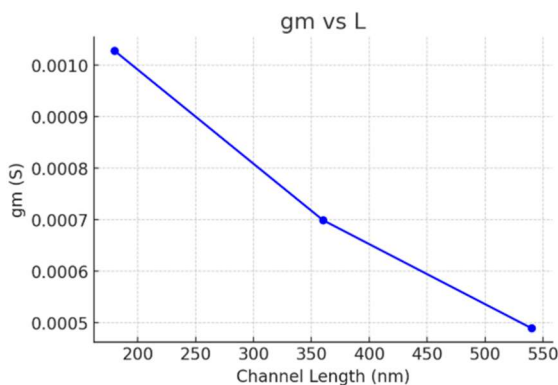
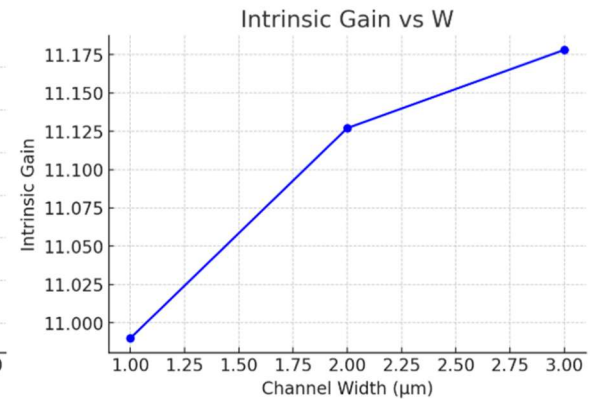
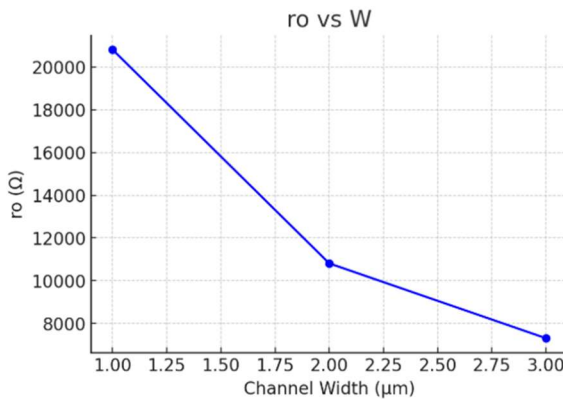
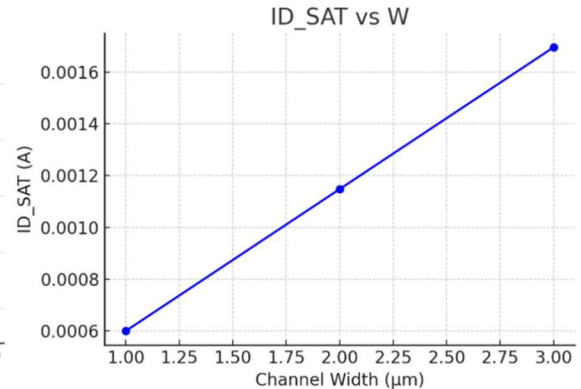
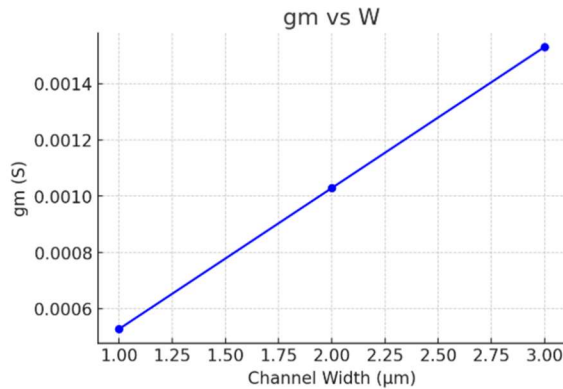
Calculations:

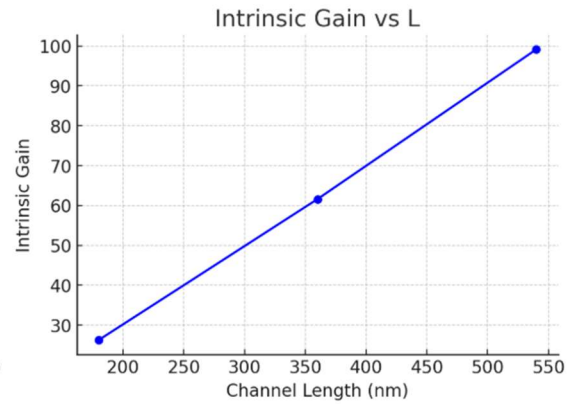
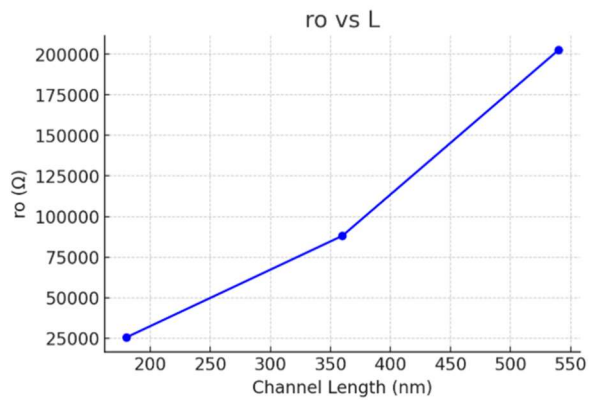
	DATE:
$W = 1 \mu m$	
$g_m = 0.000336 - 0.0007032 = 0.000527 S$	
$1.368 - 1.12858$	
$r_o = 1.06479 - 1.39686 = 22750 \Omega$	
$595 \mu A - 610 \mu A$	
$W = 2 \mu m$	
$g_m = 0.000751477 - 0.000504172 = 0.001031 S$	
$1.36 - 1.12$	
$r_o = 1.39491 - 1.06473 = 11877 \Omega$	
$1.16 mA - 1.13998 mA$	
$W = 3 \mu m$	
$g_m = 0.00109938 - 0.0007331 = 0.001496 S$	
$1.36 - 1.12$	
$r_o = 1.39491 - 1.0647 = 8009 \Omega$	
$1.7230 mA - 1.6831 mA$	
$L = 180 nm$	$r_o = 0.1798$
$0.00115473 - 0.00137$	$r_o = 10567 \Omega$
$g_m = 0.0007295 - 0.000529053 = 0.001022 S$	
0.195	
$L = 360 nm$	$g_m = 0.000704 S$
	$r_o = 16326 \Omega$
$L = 540 nm$	$g_m = 0.000497 S$
	$r_o = 20769 \Omega$
	FOR EDUCATIONAL USE

Conclusion:

In this experiment, we studied the effect of varying the aspect ratio (W/L) of an 180nm NMOS transistor on its transfer and output characteristics using Cadence Virtuoso. By systematically changing the channel width (W) and length (L) and observing their impact on the small-signal parameters (transconductance g_m and output resistance r_o), it was evident that increasing the width while keeping the length constant led to an increase in transconductance due to higher current drive capability. Conversely, increasing the channel length reduced the short-channel effects and increased the output resistance, but with a corresponding decrease in g_m . The intrinsic gain, defined as $g_m \times r_o$, was maximized for certain aspect ratios, highlighting the importance of aspect ratio optimization in analog design. The experiment reinforced the theoretical concepts of MOSFET operation and demonstrated the practical use of Cadence Virtuoso for device characterization.

Post lab Questions.





Practical Performance (5 Marks)	Presentation and Conclusion (5 Marks)	Total (10 Marks)	Sign	Date of Submission.